

Tensor Analysis

16. Relative and Absolute Tensors

A tensor $A_{r_1 \dots r_n}^{p_1 \dots p_m}$ is called a relative tensor of weight w if its components transform according to the equation

$$\bar{A}_{s_1 \dots s_n}^{q_1 \dots q_m} = \left| \frac{\partial x}{\partial \bar{x}} \right|^w A_{r_1 \dots r_n}^{p_1 \dots p_m} \frac{\partial x^{q_1}}{\partial x^{p_1}} \dots \frac{\partial \bar{x}^{q_m}}{\partial x^{p_m}} \frac{\partial x^{r_1}}{\partial \bar{x}^{s_1}} \dots \frac{\partial x^{r_n}}{\partial \bar{x}^{s_n}}$$

where $J = \left| \frac{\partial x}{\partial \bar{x}} \right|$ is the Jacobian of the transformation. If $w = 0$, the tensor is called absolute and is the type of tensor with which we have been dealing above. If $w = 1$, the relative tensor is called a tensor density. The operations of addition, multiplication, etc., of relative tensors are similar to those of absolute tensors.